

CKP Kubernetes Upgrade Guide from 1.29.0 to 1.34.3 with Calico CNI Upgrade Steps ACX

CKP Kubernetes Upgrade Guide From 1.29.0 to 1.34.3

Upgrade Plan

Phase 1: 1.29.0 → 1.30.6 → 1.31.2

Master Nodes

1.29.0 → 1.30.6 → 1.31.2

Worker Nodes

1.29.0 → 1.31.2

Phase 2: 1.31.2 → 1.32.11 → 1.33.7 → 1.34.3

Master Nodes

1.31.2 → 1.32.11 → 1.33.7 → 1.34.3

Worker Nodes

1.31.2 → 1.34.3

Calico CNI Upgrade

Upgrade Calico CNI from:

3.28.2 → 3.30.5

Note: Calico CNI upgrade will be performed only after the Kubernetes cluster is upgraded to 1.34.3 .

1. Prerequisites

1.1 Ansible Playbook

Ensure Ansible playbooks are updated to the latest version.

1.2 ETCDCTL Binary

Verify `etcdctl` binary is installed on all master nodes.

1.3 Take ETCD Backup

Run the following command on one master node:

```
1 sudo ETCDCTL_API=3 etcdctl snapshot save
2 /home/ubuntu/etcdbackuplatest.db \
3 --endpoints=https://127.0.0.1:2379 \
4 --cacert=/etc/kubernetes/pki/etcd/ca.crt \
5 --cert=/etc/kubernetes/pki/etcd/server.crt \
6 --key=/etc/kubernetes/pki/etcd/server.key
```

1.4 Validate ETCD Snapshot Size

After snapshot creation, verify snapshot file size:

```
1 ls -lrth /home/ubuntu/etcdbackuplatest.db
```

Check the live ETCD DB size:

```
1 sudo ETCDCTL_API=3 etcdctl endpoint status --write-out=table \
2 --endpoints=https://127.0.0.1:2379 \
3 --cacert=/etc/kubernetes/pki/etcd/ca.crt \
4 --cert=/etc/kubernetes/pki/etcd/server.crt \
5 --key=/etc/kubernetes/pki/etcd/server.key
```

Compare the snapshot size with the live DB size.

Both values should be identical or nearly identical.

1.5 ETCD Restore Procedure (Recovery)

In case recovery is required, run restore commands on all master nodes one by one.

Move existing ETCD directory:

```
1 mv /var/lib/etcd /var/lib/etcd-oldd
```

Restore snapshot:

```
1 ETCDCTL_API=3 etcdctl snapshot restore /tmp/etcdbackuplatest.db \
2 --data-dir /var/lib/etcd \
3 --name ckp-testing-1.novalocal \
4 --initial-cluster ckp-testing-1.novalocal=https://172.x.x.x:2380,ckp-
5 testing-2.novalocal=https://172.x.x.x:2380 \
6 --initial-cluster-token ckp-testing-1.novalocal \
7 --initial-advertise-peer-urls https://172.x.x.x:2380
```

2. Inventory Preparation

Update `inventory.ini` with the following values:

```
1 byoh_release_image=byoh-bundle-ubuntu_20.04.1_x86-64_k8s
2 cni_version=3.30.5
3 upgrade_k8s_version=1.30.6-00
```

Note: Update `upgrade_k8s_version` according to the next target Kubernetes version during each phase.

3. Pre-Upgrade Checks

3.1 Pod Statistics Baseline

Capture the current pod state for comparison after upgrade.

```
1 kubectl get pods -A -o wide | wc -l
2 kubectl get pods -A -o wide | grep -i running | wc -l
3 kubectl get pods -n <prod-ns> -o wide | grep -i running | wc -l
4 kubectl get pods -n <prod-ns> | grep -vi running
5 watch "kubectl get pods -n <prod-ns> -o wide | grep -i running"
```

3.2 Database Health Check

Verify all DB clusters are healthy before proceeding.

3.3 DB Leader Switchover (Before Drain)

Ensure:

- DB leader is moved to another node
- Replication is healthy before proceeding with node drain

4. Upgrade Execution

4.1 Upgrade Master Nodes

Run master node upgrade playbook:

```
1 ansible-playbook -i inventory.ini playbooks/Upgrade-k8s-mastres.yaml -v
```

4.2 Upgrade Worker Nodes

NOTE: If worker node count is high, it is recommended to upgrade worker nodes in batches of 4–5 nodes.

Before draining each worker node, update the worker upgrade role.

Edit:

```
1 vi roles/upgrade-k8s-workers/tasks/main.yaml
```

Add the following section immediately before the worker drain task:

```
1 - name: Confirm before draining node
2   pause:
3     prompt: "Do you want to proceed with drain of worker node {{
4     node_name.stdout }}? (yes/no)"
5     register: drain_confirmation
6     delegate_to: localhost
```

Run worker node upgrade playbook:

```
1 ansible-playbook -i inventory.ini playbooks/Upgrade-k8s-workers.yaml -v
```

4.3 Upgrade Calico CNI

Ensure `jq` is Installed

Verify that `jq` is installed on the control node and all required cluster nodes before starting the CNI upgrade.

Check:

```
1 | jq --version
```

If not installed:

Ubuntu/Debian

```
1 | sudo apt update
2 | sudo apt install -y jq
```

Edit the Calico upgrade role:

```
1 | vi roles/calico-cni-upgrade/tasks/main.yaml
```

Update the CRD apply shell section:

```
1 | shell: "kubectl apply --server-side --force-conflicts -f
https://raw.githubusercontent.com/projectcalico/calico/v{{ cni_version
}}/manifests/operator-crd.yaml"
```

Run Calico CNI upgrade playbook:

```
1 | ansible-playbook -i inventory.ini playbooks/calico_cni_upgrade.yaml -v
```

4.4 CoreDNS ConfigMap Validation

After Calico/CoreDNS upgrade activity, validate that the CoreDNS configuration is identical.

```
1 | kubectl get configmap coredns -n kube-system -o yaml > coredns.yaml
2 | kubectl get configmap coredns-new -n kube-system -o yaml > coredns-
new.yaml
3 |
4 | diff -u coredns.yaml coredns-new.yaml
```

Ensure both ConfigMaps are identical before proceeding.

5. Post-Upgrade Validation

5.1 Cluster Health Validation

Verify:

- All nodes are in `Ready` state
- All pods are healthy
- No `CrashLoopBackOff` containers

- All application namespaces are healthy

Useful commands:

```
1 kubectl get nodes
2 kubectl get pods -A
3 kubectl get pods -A | grep -i crash
4 kubectl get events -A --sort-by=.metadata.creationTimestamp
```

5.2 DNS Validation

Validate private DNS lookup through cluster DNS service:

```
1 nslookup mip-dev-ws.rumadm.net 10.96.0.10
```

Ensure DNS resolution succeeds correctly.

```
1 kubectl run busybox-test --image=busybox:1.36 -it --rm --restart=Never
-- nslookup ruauth.metro-cc.ru
```